Examiner only

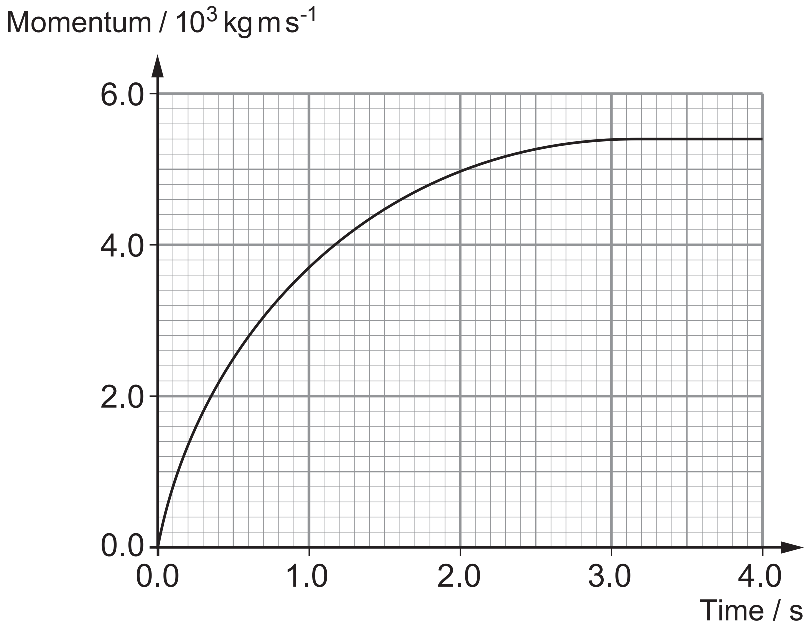
5. *(a)* A law of motion can be expressed as:

$$\text{resultant force} = \frac{\text{change in momentum}}{\text{time}}$$

State the name of the law. [1]

....

(b) The graph shows how the momentum of a spacecraft varies with time.



(i) By drawing a suitable tangent, show that the resultant force on the spacecraft at t = 1.0 s is approximately 2 kN. [2]

....

....

....

(ii) Hence show that the mass of the spacecraft is approximately 5 000 kg, given that its acceleration at t = 1.0 s is 0.4 m s^{-2}. [1]

....

....

(iii) **Label, with the letter P**, a point on the graph where the resultant force on the spacecraft is zero. [1]

Examiner only

(c) At t = 4.0 s the spacecraft 'docks' (collides) with another **stationary** spacecraft of mass 7 000 kg. They join on impact.

(i) State the principle of conservation of momentum. [2]

....

....

....

(ii) Calculate the velocity of both spacecraft after colliding. [3]

....

....

....

....

....

10